

Data Analysis Final Assignment Report

Team: Analog Avengers
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1 Contributions

- Eingang Fabian: Dataset selection and acquisition, Data quality analysis and preprocessing pipeline
- Kotschnig Thomas: Visualizations and EDA, Probability analysis tasks
- Krenn Matthias: Regression modeling and interpretation, Report writing and figure polishing

2 Dataset Description

- "Bike sales in Europe" from <https://www.kaggle.com/datasets/sadiqshah/bike-sales-in-europe>
- It has more than 100k entries of sales data from different countries. Stretching from 2011 to 2016, with a daily sampling frequency.
- Key variables analyzed: customer age, order quantity, unit cost, unit price, profit, cost, revenue
- Shape: 113036 rows x 18 columns
- No missing data, however, the entry of some dates is missing completely. This resolves in no missing data, but inconsistent time series. There is only one bigger gap, therefore we decided for it to be okay.

3 Task 1. Data Preprocessing and Basic Analysis

3.1 Basic statistical analysis using pandas

Statistical summary of key numeric variables was obtained using pandas `describe()` function:

Table 1: Descriptive statistics

	Customer Age	Order Qty	Unit Cost	Unit Price	Profit	Cost	Revenue
Count	113036	113036	113036	113036	113036	113036	113036
Mean	35.92	11.90	267.30	452.94	285.05	469.32	754.37
Std	11.02	9.56	549.84	922.07	453.89	884.87	1309.09
Min	17	1	1	2	-30	1	2
25%	28	2	2	5	29	28	63
50%	35	10	9	24	101	108	223
75%	43	20	42	70	358	432	800
Max	87	32	2171	3578	15096	42978	58074

Table 2: Grouped summary of Revenue, Profit, and Order Quantity by Country

Country	Revenue			Profit		Order Quantity	
	Sum	Mean	Count	Sum	Mean	Sum	Mean
United States	27975547	713.55	39206	11073644	282.45	477539	12.18
Australia	21302059	889.96	23936	6776030	283.09	263585	11.01
United Kingdom	10646196	781.66	13620	4413853	324.07	157218	11.54
Germany	8978596	809.03	11098	3359995	302.76	125720	11.33
France	8432872	766.76	10998	2880282	261.89	128995	11.73
Canada	7935738	559.72	14178	3717296	262.19	192259	13.56

3.2 Original data quality analysis including visualization

There are no missing data in our dataset. This is why we did not add any visualization of this parameter. However, there is a timeline gap visible in the figure below.

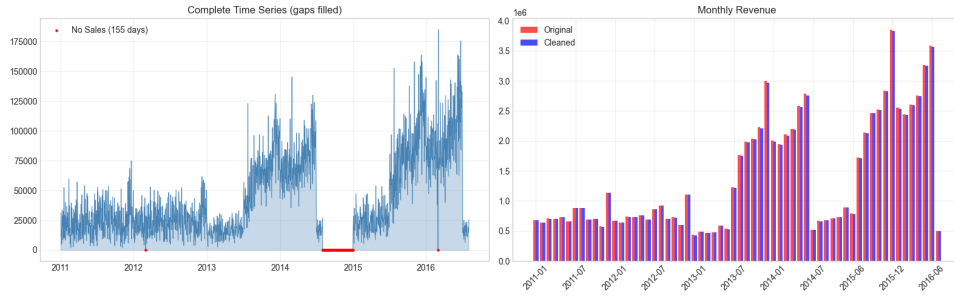


Figure 1: Consistency checks in time series and comparison of revenue before and after preprocessing

Outliers have been identified via IQR method seen in the figure below, but they have not been removed. This is because we wanted this data in the dataset for better and full analysis. Otherwise the dataset would have lost too much information.



Figure 2: IQR applied on some key variables to identify outliers.

3.3 Data preprocessing

- Cleaning steps performed: Duplicates have been dropped.
- Missing-value treatment: No missing values were present in the dataset, therefore no treatment was necessary.
- Outlier handling: Outliers identified via IQR method ($1.5 \times \text{IQR threshold}$) but intentionally retained. Justification: These extreme values contain valuable business insights (high-value transactions, unusual market events) that would be lost if removed.

- Feature engineering:
Time-based features added: DayOfWeek, DayOfWeek_Name, WeekOfYear, Quarter, IsWeek-end
Financial features created: Profit_Margin, Avg_Unit_Profit, High-value flag
- Final dataset shape after preprocessing: (8 new features added, no rows removed)
Original 113,036 rows $\times$ 18 columns $\rightarrow$ Cleaned 113,036 rows $\times$ 26 columns

3.4 Preprocessed vs original data visual analysis

The accuracy of the dataset improved by dropping the duplicates. A trade-off would be the possible removal of relevant data (no real duplicate).

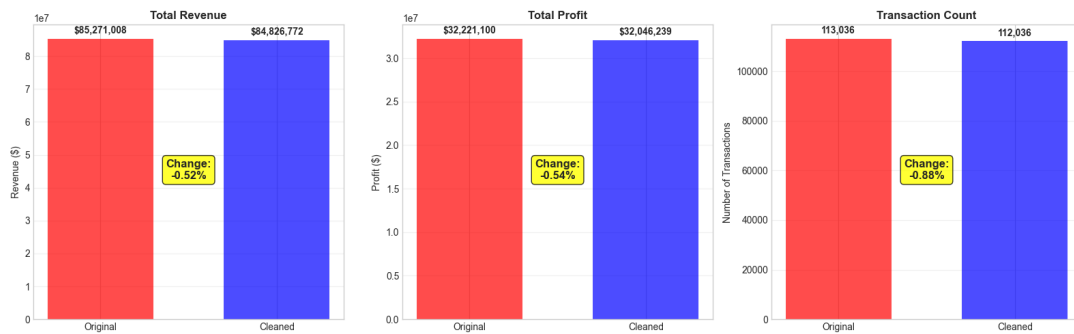


Figure 3: Comparison of key sales values due to the removal of 1000 duplicates.

4 Task 2. Visualization and Exploratory Analysis

4.1 Time series visualizations

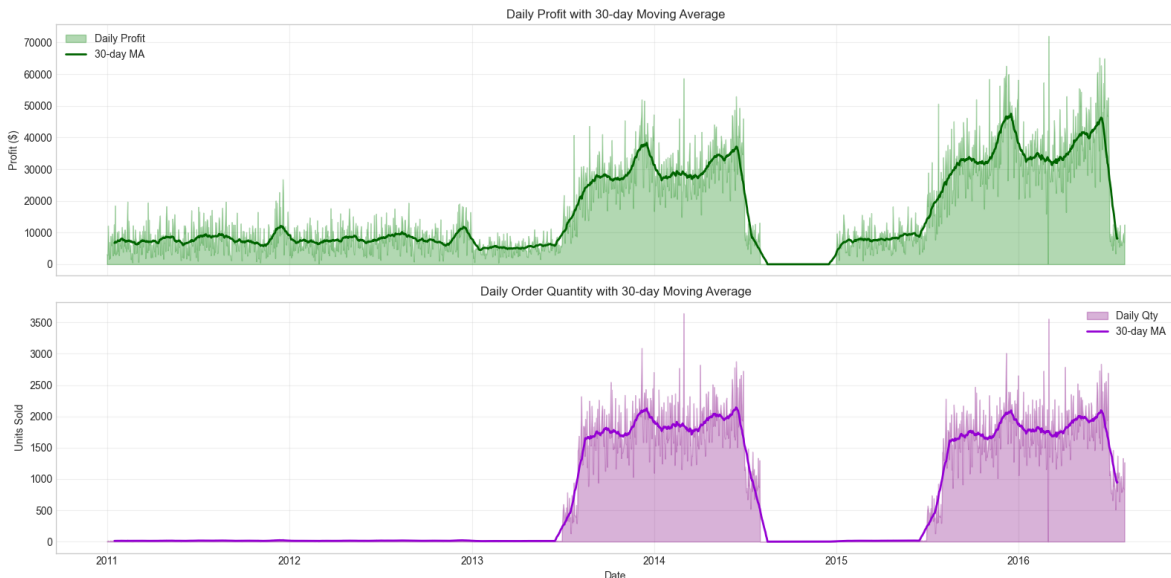


Figure 4: Time series visualization of sales over time.

4.2 Distribution analysis with histograms

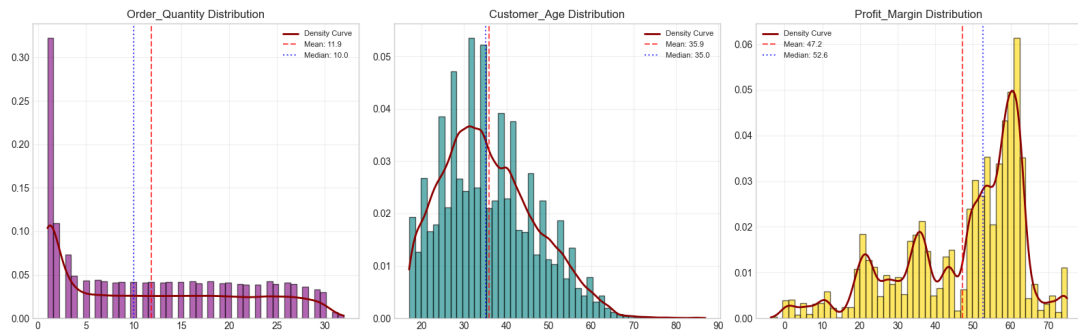


Figure 5: Histograms of key numeric variables showing distribution shapes.

Order Quantity: Nearly symmetric distribution (skewness: 0.378) with light tails, indicating values are concentrated near the center with fewer extreme outliers. The bimodal pattern suggests two distinct purchasing behaviors.

Customer Age: Right-skewed distribution (skewness: 0.524) with normal-like tails. The multimodal pattern reflects age clustering across different customer segments. Mean age of 35.92 years exceeds the median (35.00), confirming right skew with older customers in the tail.

Profit Margin: Left-skewed distribution (skewness: -0.856) with normal-like tails, indicating most transactions cluster at higher profit margins with a tail toward lower margins. The multimodal pattern suggests distinct profit tiers based on product categories.

4.3 Correlation analysis and heatmaps

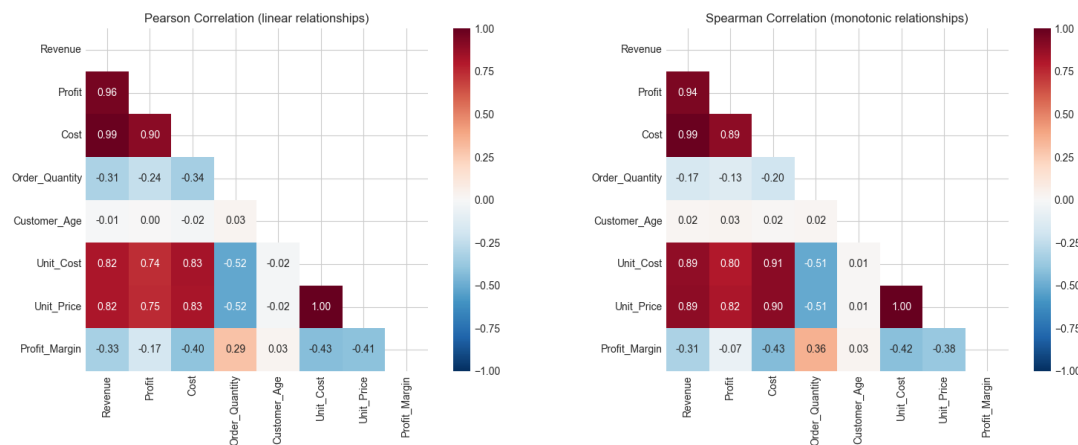


Figure 6: Correlation heatmap of key numeric variables.

Both correlation types have been calculated. Both types show the same strongest correlations. Those correlations are between the financial variables revenue, cost and profit. That makes sense, due to higher revenue leading to higher profit. Or higher costs also lead to higher revenue.

4.4 Monthly pattern analysis

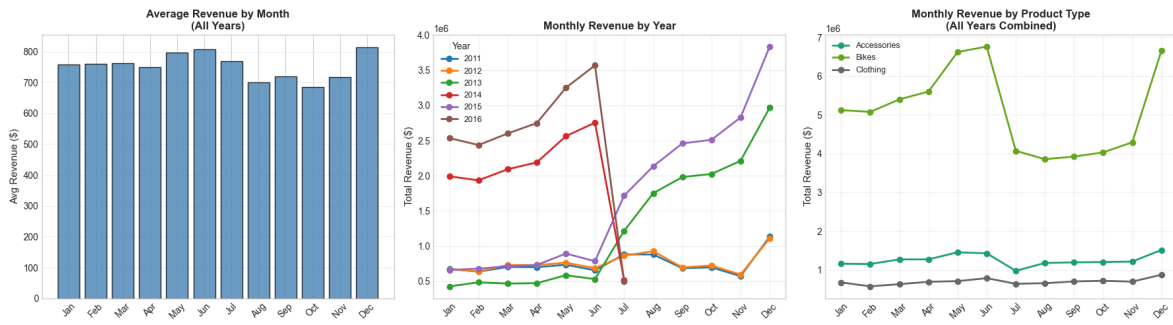


Figure 7: Monthly pattern analysis of sales.

Our dataset doesn't provide any timestamp data for the sales. Therefore, it was grouped into monthly data and reviewed over the whole year. The data of every month was then averaged over the years. It is very interesting to see the strong seasonal patterns of bike sales. Our dataset shows almost identical sales on the weekend compared to the weekdays. This shows that the shop was also open on weekends. Which is very unusual for European shops. Therefore, it has to be an online shop. It was interesting to see a deviation in customer age depending on the day of the week.

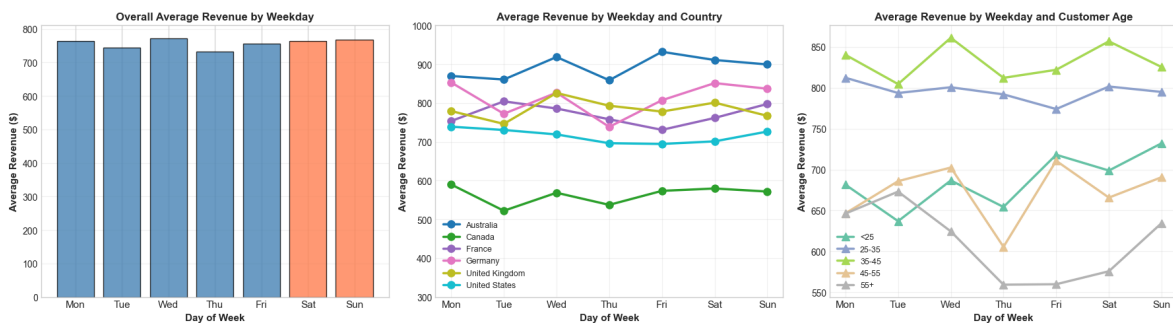


Figure 8: Weekday pattern analysis of sales.

4.5 Summary of observed patterns, similar to True/False questions

- Revenue shows a positive long-term trend over the dataset period TRUE
 - Evidence: 90-day MA grew 308.2% from start to end
- Q4 (Oct-Dec) shows significantly higher sales than other quarters FALSE
 - Evidence: Q4 revenue is not higher than the average of other quarters
- Revenue and Profit are strongly positively correlated ($r > 0.8$) TRUE
 - Evidence: Pearson $r = 0.957$
- Customer age has minimal impact on transaction revenue TRUE
 - Evidence: Age-Revenue correlation $r = -0.009$
- December shows the highest monthly average revenue TRUE
 - Evidence: Best month: Dec, worst: Oct

5 Task 3. Probability Analysis

5.1 Threshold-based probability estimation

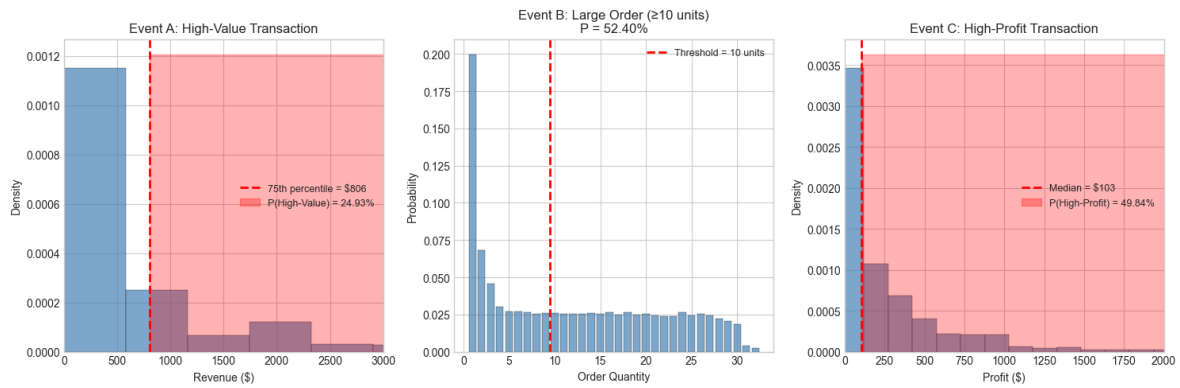


Figure 9: Threshold-based probability estimation visualization.

The three thresholds chosen are:

- High value transaction $\rightarrow$ 75% was used as threshold
- Order quantity $\rightarrow$ everything above 10 units was considered high
- High profit transaction $\rightarrow$ mean profit value was used as threshold

5.2 Cross tabulation analysis

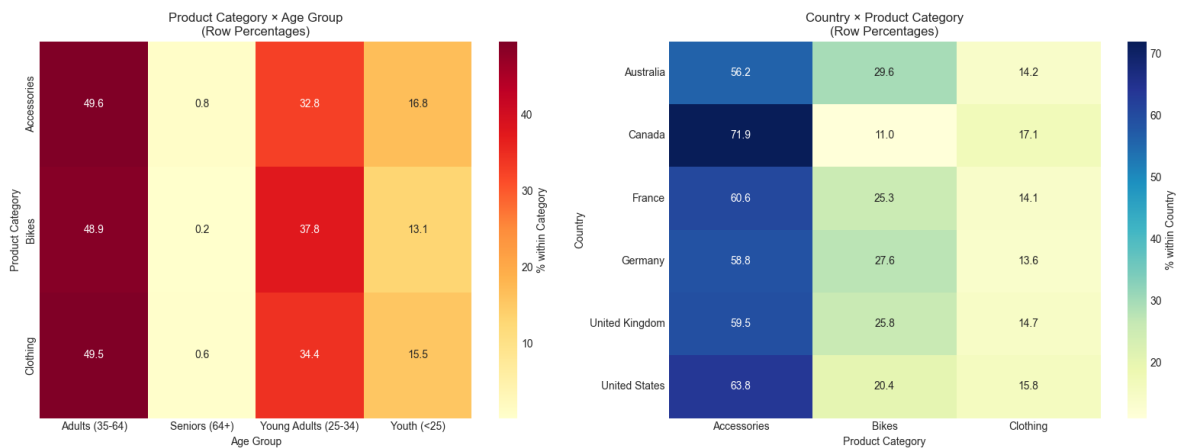


Figure 10: Cross tabulation analysis visualization.

It is interesting to see that the product category is not not influenced by the customer age. However, in the right cross tab it is visible that in Canada the amount of bike sales is much lower than in the rest of the countries. But the accessories are sold way more often in Canada than in the other countries.

5.3 Conditional probability analysis

$$P(\text{High-Value}) = 24,93\%$$

$$P(\text{High-Value} \mid \text{Bikes}) = 80,79\%$$

$$P(\text{High-Value} \mid \text{Canada}) = 17,92\%$$

$$P(\text{High-Profit}) = 49,84\%$$

$$P(\text{High-Profit} \mid \text{Large Order}) = 44,93\%$$

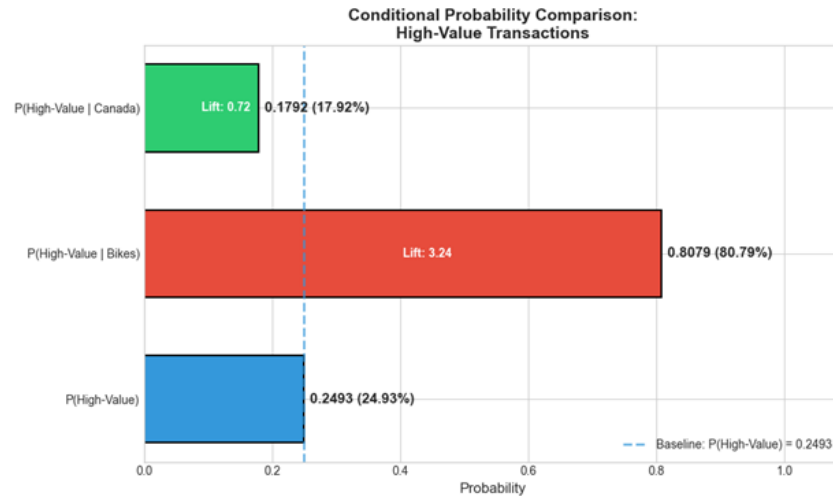


Figure 11: Conditional probability analysis visualization.

When a Bike is sold the probability that the Transaction is a high Value transaction increases with the Factor of 3.24. When we compare the probability that a Transaction is a high profit transaction to the probability that a transaction is a high profit transaction under the condition that the order quantity is flagged with large order, we can see a decrease in the Probability by the Factor 0.72.

5.4 Summary of observations from each probability task

- Key takeaway from threshold probability: Choosing thresholds requires care—too high or low can misrepresent risk; advantage is simplicity and clear segmentation, but it ignores interactions beyond the set cutoffs.
- Key takeaway from crosstab: Useful for detecting dependencies between categorical variables, but small sample sizes can give misleading p-values; it's easy to interpret but limited to pairwise relationships.
- Key takeaway from conditional probability: Shows how one event affects another, highlighting trends; risk lies in misinterpreting correlation as causation, though it provides actionable insights for targeting specific segments.

6 Task 4. Statistical Theory Applications

6.1 Law of Large Numbers (LLN) demonstration

For the analysis the revenue was chosen, because it make a lot of sense for financial analysis. Knowing the mean revenue of 100 sales for example and how fast is converges to the sample mean n . It is visible that the sample mean converges to the population mean with increasing sample size.

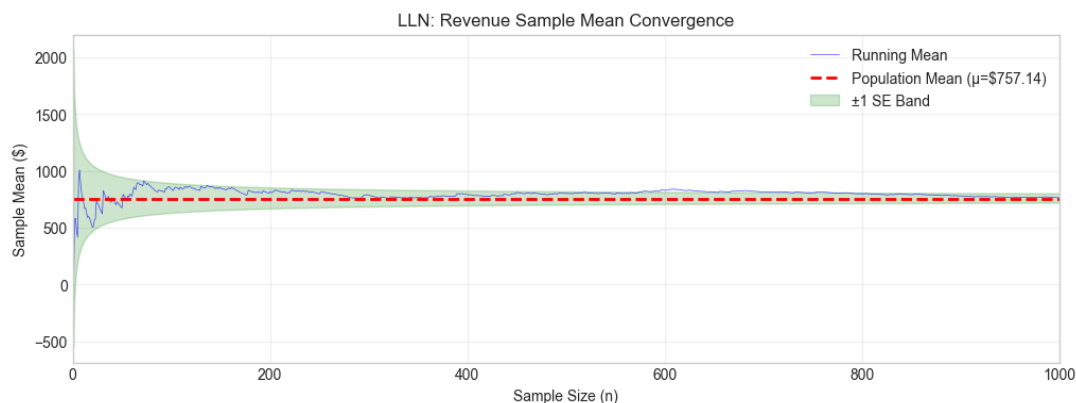


Figure 12: LLN demonstration visualization.

6.2 Central Limit Theorem (CLT) application

We used different sample sizes to show the CLT. Ranging from 10 to 500. For every sample size 2000 number of trials were performed with replacement.

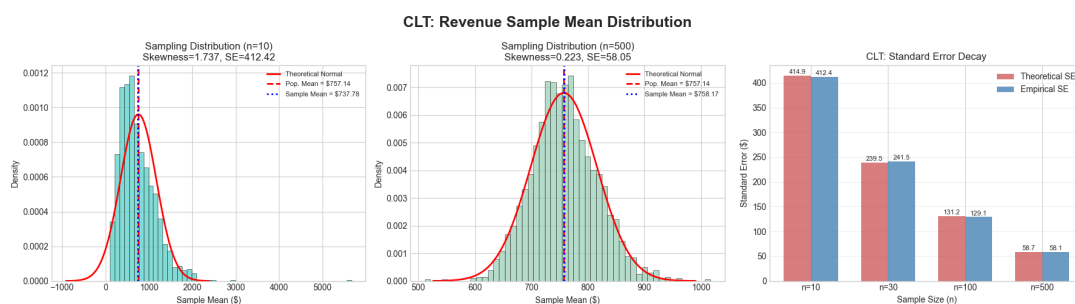


Figure 13: CLT demonstration visualization.

6.3 Result interpretation

- What LLN showed in your data context:
That the sample mean of the revenue can be used as an mean value for the average revenue per sale.
- What CLT showed, and any deviations and why:
The Central Limit Theorem is demonstrated by the sampling distributions of the sample mean becoming increasingly normal as sample size increases. For smaller sample sizes, slight skewness is observed due to the underlying revenue distribution, but this deviation decreases for larger samples. The standard error of the sample mean decreases at the theoretical rate $\sigma/\sqrt{n}$, closely matching empirical estimates.

7 Task 5. Regression Analysis

7.1 Linear or Polynomial model selection

- Define target y and predictors X : Target y is daily revenue, chosen as the key business KPI that directly informs budgeting and planning decisions. Predictors X include volume metrics (Daily_Qty, Daily_Cost), temporal features (DayOfWeek, Month, IsWeekend), customer demographics (Avg_Age), and categorical variables (Top_Category, Top_Country) to capture market segment effects.
- Motivation for linear vs polynomial: Linear regression serves as an interpretable baseline where coefficients directly quantify each feature's impact on revenue. Polynomial regression (degree 2) tests whether non-linear relationships or feature interactions (e.g., quantity $\times$ category effects) improve predictions. Model selection prioritizes test R^2 while monitoring the overfitting gap (train R^2 - test R^2); the simpler model is preferred when performance is comparable.
- Any train-test split rationale (time-aware split if relevant): A time-based 80/20 split was used, training on earlier dates (2011–2015) and testing on recent data (2015–2016). This respects temporal ordering, prevents data leakage from future observations, and simulates a realistic forecasting scenario.

7.2 Model fitting and validation

- Fit procedure and preprocessing (scaling, feature selection): StandardScaler applied to all numeric features for stable gradient descent and comparable coefficient magnitudes. Categorical features (Top_Category, Top_Country) were one-hot encoded with the first category dropped to avoid multicollinearity.
- Validation method (holdout, time-series split, etc.): Time-based holdout split ensures the model is evaluated on genuinely unseen future data. This is critical for time-series regression to avoid overly optimistic performance estimates.
- Metrics reported (RMSE, MAE, R^2) and why:
 - R^2 : Proportion of variance explained, primary criterion for model selection
 - RMSE: Root mean squared error penalizes large deviations, reported in dollars
 - MAE: Mean absolute error provides a robust, interpretable average prediction error

Results: Linear Regression achieved Test $R^2 = 0.9954$, RMSE = \$2,267, MAE = \$1,808. Polynomial Regression achieved Test $R^2 = 0.9956$, RMSE = \$2,227, MAE = \$1,784, with overfitting gap of 0.0017. Polynomial regression (degree 2) was selected due to marginally better test performance.

- Residual analysis (at least one plot recommended): Four diagnostic plots were created (Figure ??):
 - Residuals vs Predicted: Random scatter centered around zero with no funnel shape
 - Q-Q Plot: Points closely follow the theoretical line with minor deviations at the tails, confirming approximate normality of residuals.
 - Residuals Over Time: No systematic trend or drift visible; variance appears stable across the test period.
 - Actual vs Predicted: Points tightly clustered along the diagonal, demonstrating excellent predictive accuracy.

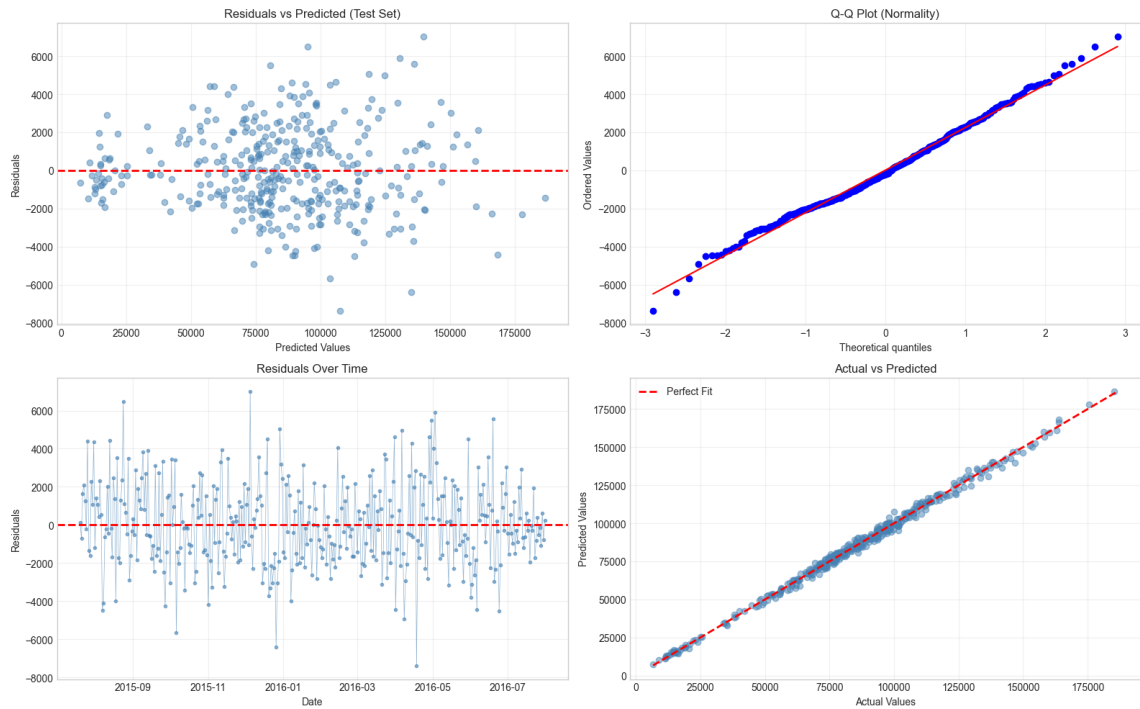


Figure 14: Residual diagnostics for the polynomial regression model (degree 2).

7.3 Result interpretation and analysis

- Main effects and practical meaning: Daily_Cost and Daily_Qty are the dominant predictors, which aligns with business logic, more items sold at higher cost directly increases revenue. The polynomial model (degree 2) captures subtle non-linear interactions between features while maintaining excellent generalization.
- Failure cases or where model performs poorly: Higher polynomial degrees (3 and 4) exhibit catastrophic overfitting, with test R^2 dropping to extreme negative values (see Figure ??). This confirms that degree 2 is optimal, complex enough to capture interactions but not so flexible that it memorizes training noise. Remaining prediction errors occur on days with unusual sales patterns not explained by the available features.

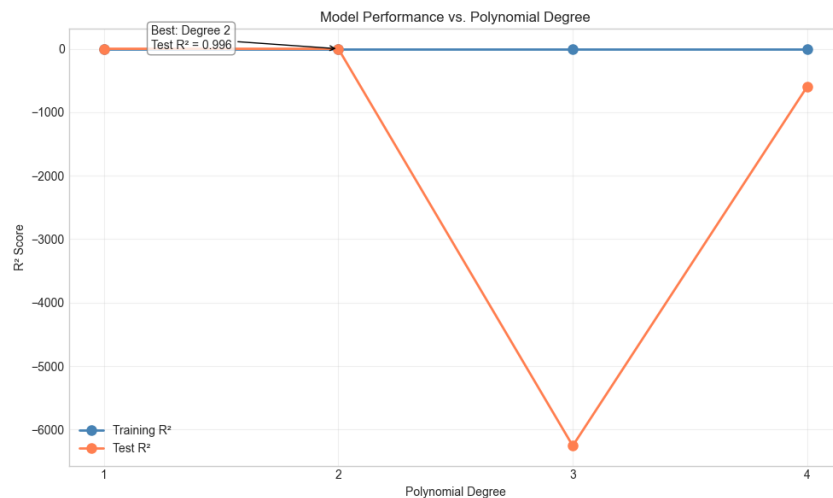


Figure 15: Model performance vs polynomial degree. Degrees 3–4 show severe overfitting.

8 Task 6. Dimensionality Reduction and Statistical Tests

8.1 Dimensionality Reduction

Three dimensionality reduction techniques were applied to visualize the high-dimensional sales data (Revenue, Profit, Cost, Order_Quantity, Unit_Cost, Unit_Price, Customer_Age, Profit_Margin) and examine whether product categories form natural clusters.

PCA Analysis (Figure ??):

- Scree plot shows PC1 explains $\sim 60\%$ of variance, with PC1–PC2 together capturing $\sim 74\%$.
- Feature loadings reveal that Profit_Margin dominates PC1, while Customer_Age drives PC2. Financial variables (Revenue, Profit, Cost) load similarly on PC3.
- PCA projection shows partial separation of Bikes (orange) from Accessories/Clothing, but the categories overlap substantially in linear space.

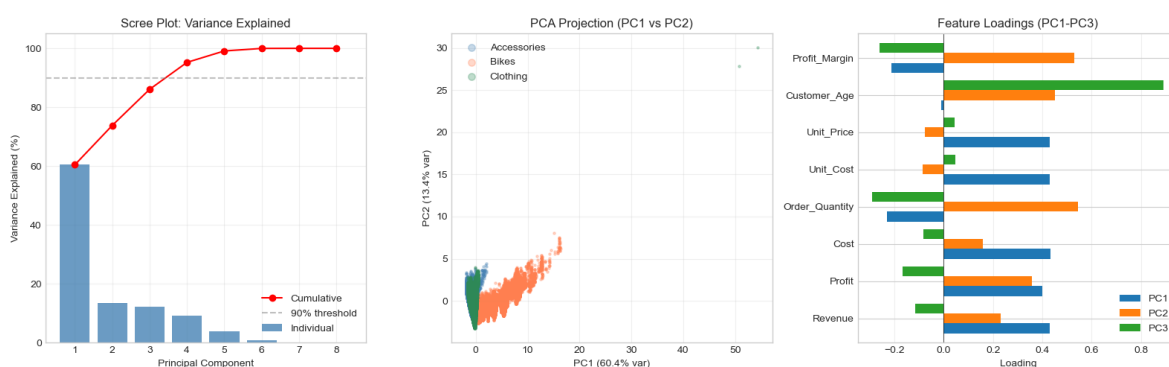


Figure 16: Variance explained, PCA projection, and feature loadings for PC1–PC3.

Method Comparison (Figure ??):

- PCA (linear): Shows global structure but the main cluster appears compressed due to outliers stretching the axis scale. Categories overlap substantially.
- t-SNE and UMAP: Both non-linear methods produce similar results and show clearer separation between categories compared to PCA. Bikes (orange) form a distinct cluster, while Accessories and Clothing show partial overlap but are still distinguishable. This suggests non-linear methods better capture the underlying structure in the data.

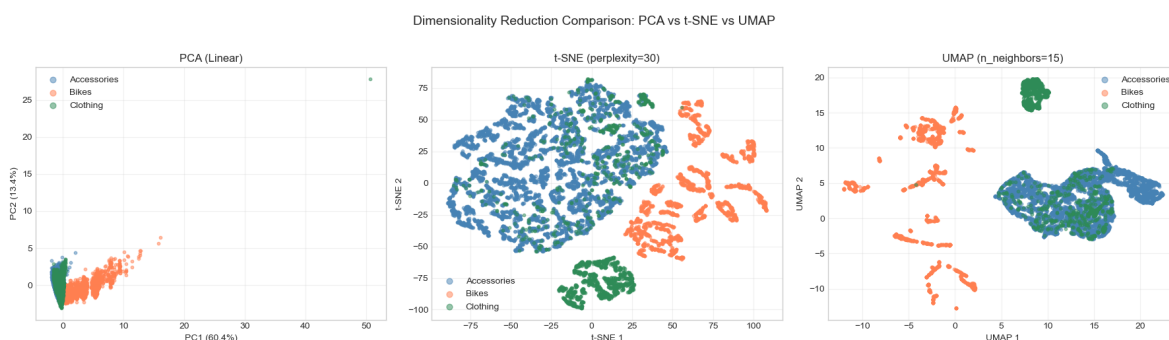


Figure 17: Comparison of PCA, t-SNE, and UMAP projections colored by product category.

8.2 Result interpretation and analysis

- Main effects and practical meaning: Bikes are distinguishable from Accessories and Clothing primarily due to higher unit prices and profit margins. However, Accessories and Clothing overlap significantly, suggesting similar purchasing patterns.
- Failure cases or where model performs poorly: Linear methods (PCA) struggle to separate categories because the numeric features alone do not fully encode product type. Non-linear methods (UMAP) perform better but still show overlap, indicating that categorical labels provide information beyond what numeric features capture.

9 Key Findings and Conclusions

Preprocessing and EDA:

The dataset was of high quality, with no missing values and only minor temporal gaps. Duplicate removal improved consistency with minimal impact on overall trends. Exploratory analysis revealed strong seasonality, a clear positive long-term revenue trend, and very strong correlations among financial variables (revenue, cost, profit). Customer age showed negligible influence on revenue, indicating that sales performance is primarily driven by product and volume rather than demographics.

Probability Analysis:

High-value transactions account for roughly one quarter of all sales, but the probability increases by more than a factor of three when conditioning on bike sales. Large order quantities do not necessarily imply higher profit probability, suggesting potential margin trade-offs. Geographic differences were observed, particularly Canada's relatively higher share of accessory sales compared to bikes.

LLN and CLT:

The Law of Large Numbers was confirmed as sample mean revenue converged to the population mean with increasing sample size. The Central Limit Theorem was empirically validated, with sampling distributions of the mean approaching normality for larger samples despite skewness in the underlying revenue distribution.

Regression Analysis:

Daily revenue was predicted with very high accuracy using cost, quantity, temporal, and categorical features. A degree-2 polynomial regression slightly outperformed the linear model while maintaining strong generalization, whereas higher-degree models exhibited severe overfitting. Residual diagnostics showed no systematic bias or temporal drift.

Limitations and Future Work:

The absence of detailed timestamps and richer customer information limits finer-grained analysis. Outliers influence variance and dimensionality reduction results. Future work would focus on additional feature engineering, time-series models, customer segmentation, and the inclusion of external factors to improve explanatory power.

What you would do next if you had more time:

We would investigate additional features that could improve the data analysis. Such as our feature engineering steps already showed, that new features can add a lot of value. We would investigate those new features and their impact on the analysis.

10 Reproducibility Notes

- Exact dataset source link, last accessed on 28.01.2026: <https://www.kaggle.com/datasets/sadiqshah/bike-sales-in-europe>
- Key libraries used and versions:
 - Python 3.10.11
 - pandas 2.3.3
 - numpy 2.2.6
 - matplotlib 3.10.8
 - seaborn 0.13.2
 - scipy 1.15.3
 - scikit-learn 1.3.2
 - plotly 5.16.0
 - umap-learn 0.5.4
- How to run the notebook end-to-end:

For running the notebook error free, we have created a “Install.bat” batch file. The batch file installs:

 - The recommended Python version for using the notebook.
 - The necessary packages.

Important prerequisite:

A windows system.

Step-by-Step user guide:

1. Clone the github repository: ‘git clone <https://github.com/Kotschnig-edu/DataAnalysis.git>’
2. Run the “Install.bat” file in the cloned repository.
3. Open the notebook and select the created python virtual environment called “.venv”
4. Run the cells of the notebook in the sequential order or press “Run all” in your programming environment.